ISYS2120 Assignment 2 (Conceptual Model) - Sem2, 2025

Due: Sunday 21 September, 11:59pm Sydney time

Value: 5% for group aspect, and 7% for individual aspect

Late work policy: (from unit outline) Late submissions for assignments will incur a penalty of 5% of the maximum awardable marks for each day, or part-day, past the due date, up to a maximum of 7 days (as after this time, feedback on on-time submissions will be available, resulting in an unfair advantage if submissions after this time were accepted). After 7 days late submissions will not be accepted. Where special consideration is granted for these assessments, extensions of a maximum of 7 days will be permitted. After 7 days, reweighting of other relevant tasks will be applied.

Summary (but details further down are authoritative)
Groups: with 2 or 3 people from a single lab session, who all get the same mark from the group submission (except that if some member is considered to have not contributed reasonably, then the unit coordinator may reduce their score appropriately.)
Provided for you: a textual description about a domain.

You produce: as a group, a conceptual model for the data to be stored about the domain, expressed in an extended ER diagram; also a discussion of any issues where the text was unclear and you had to make a decision for the model, and overall strengths and weakness of the model.

Also, each member as an individual produces: a description of how you checked the validity and completeness of (some version, perhaps preliminary, of) the group's conceptual model; and a statement of the process the group followed to produce its conceptual model.
Submit (as a group): a report with structure as described in detail below [submit on Canvas]

Submit (as an individual): a report with structure as described in detail below [submit on Canvas]

Group Formation and Policy: This assignment is done in small groups of 2 or 3 students. We ask that all students in a group be coming to the same lab session, so

you can work together more easily and the tutor can track what is happening. Groups are formed in week 4 lab session but may be adjusted by tutor or coordinator when necessary (e.g. to include someone who was away in week 4).

Procedure: There will be a Canvas group for this assessment and the teaching team will allocate students as members in the appropriate group. If the group shown in Canvas doesn't match what you believe was agreed, Post privately on Ed urgently. If you are not in a Canvas group, post privately on Ed as soon as possible, because you will get no group aspect mark unless you are in a Canvas group.

If, during the course of the assignment work, there is a dispute among group members that you can't resolve, or that will impact your group's capacity to complete the task well, you need to inform the unit coordinator nasim.ahmed@sydney.edu.au. Make sure that your email names the group and is explicit about the difficulty; also make sure this email is copied to all the members of the group. We need to know about problems in time to help fix them, so set early deadlines for group members, and deal with non-performance promptly (don't wait till a few days before the work is due, to complain that someone is not doing their share). If necessary, the coordinator will split a group and leave anyone who doesn't participate effectively in a group by themself. 48 hours before the deadline, we will stop taking complaints anything related to the group conflict.

The task:

The Western Sydney Airport (WSA) is a new world-class international airport currently under construction, built to service 10 million passengers a year with the first stage expected to be open by December 2026. They are hiring you to guide the data management which will be needed. Your first task is to model the data which will be kept in order to deal with aircrafts that will be stationed and maintained at the airport, as well as the workers.

The airport has a technical department that maintains aircrafts. The new information system shall support the maintenance units of the technical department by storing information about every aircraft maintained at the airport. The aircrafts are identified with a unique international registration number. Each registration number consists of a unique two-letter code identifying the aircraft owner and a four-letter unique id within the airline fleet. For example, 'QF-APAC'

would be a valid aircraft registration number. 'QF' is the code for the Qantas as the owner of the fleet of aircrafts and 'APAC' identifies one particular aircraft of the Qantas fleet. Furthermore, airline often associates a specific name to an aircraft. For example, 'QF-APAC' is named 'Gadigal Land' by Qantas. Each time a plane is held at the airport for maintenances, the airport will record the dates of arrival and departure and which airport the plane has come from.

Each airline must have a contact phone, email, and a registered website. Several airlines can be owned by the same company. Each airline owning company is identified by a unique name, and associated to a home country. Every aircraft has a specific type and each aircraft type has a unique model name (e.g., 'A380-800') and is produced by a specific maker (e.g., Airbus). The airport will keep track of a number of technical details for each aircraft type such as number of engines, maximum number of passengers, maximum cruising speed, as well as aircraft length, span and height.

When it comes to the human resources, the airport's initial focus is on three important departments as they recruit for their respective staff: ground crew, technicians, and traffic controllers. Each employee must have their name and salary captured. They are uniquely identified by a tax file number (TFN). Employees can join an airport-supported social club. The system will keep a unique membership number for those enrolled employees.

The ground crews ensure the safety and comfort of passengers. Employees may occasionally be called upon to work overtime. Should that occur, the extra hours must be entered on each day so that the crews can be compensated accordingly. We assume that each technician is an expert in at least one type of aircraft, but not more than a total of six. The airport records the technician's name, TFN, address, mobile number, email, and salary. The technician's level of proficiency should also be tracked for each type of aircraft they work on. For traffic controllers, a medical examination must be completed every year and the most recent examination date needs to be logged, as well as the doctor who performed the examination.

The airport has a number of compliance test specifications that are periodically performed to ensure that aircrafts are still airworthy. Each test specification has a Civil Aviation Safety Authority (CASA) test number, name, passing score, and maximum total score. The CASA test number is a unique serial number which

clearly identifies each test specification. The CASA requires the airport to keep track of each time a given aircraft is tested by a technician with regards to a certain test specification. Each testing event records the date and time, the number of hours the technician spent on the test, and the score the aircraft received on the test.

Note: you may find this description lacks important details; requests for clarification should be posted on the unit Edstem discussion board, under the posting category of Assignment 2, You should observe closely any answers posted by staff (not only to your own questions, but also to questions from others. Also please be mindful of asking repeated questions, please check if the same question has been asked already, before posting on Ed.)

Expectations of weekly progress (to get feedback from tutor)

The tutor will be available for consultations about your progress on the assignment in the hour after the tutorial.

***Week 5 progress*:** each member should have read the instructions, and make sure that you understand what is expected. You may use the consultation time to ask the tutor for anything that is unclear.

***Week 6 progress*:** each group should have a proposal for at least four entity sets, and some relationships between them; including proposals for the attributes of these entities, and constraints (such as primary keys, and any multiplicity aspects of the relationships chosen). We encourage you to show this design to the tutor for feedback in the consultation time.

***Week 7 progress*:** each group should have an ER diagram which should be close to final. We encourage you to discuss this with the tutor during their consultation hour.

Submission (group)

As a group, you must produce a report with the following two-part structure. The report can be produced with whatever document formatting approach you choose, but what is uploaded must be in PDF. One member of the group should upload this to the canvas submission link, before the due date. The report should begin with a cover sheet, which follows the template we provide, and lists the members of the group who produced this report together.

Part A: An extended ER diagram that captures a conceptual data model for this domain. It is acceptable to have a single diagram, or to give a high-level diagram that omits the attributes, along with detailed diagrams for each entity and relationship, where the attributes are shown.

Part B: It's important to remember that real-world descriptions, like the one provided by us, often come with ambiguities or missing details, as the real world can be complex and open to interpretation. In your modelling process, you might encounter situations where the provided description doesn't explicitly state how something should be represented. In Part B, provide an explanation, of any aspects of the domain where your model (as presented in Part A) reflects a decision that was not given directly in the textual description. This includes aspects where staff clarified the intention on Edstem, in such cases, Part B should refer to the Edstem thread on which you relied in the model. For example, if you decided to make some attribute multi-valued, when this wasn't stated for you, or if you chose an attribute to be a primary key when it was not already described as an identifier. Also, discuss strengths and limitations of your conceptual model as a reflection of the domain. In particular, state explicitly any aspects or constraints of the domain (as given in text or Edstem answers by staff) that are not captured in your conceptual model, and say why these were not in your model.

Submission (individual)

Each member must produce a **report** with the following three-part structure. The report can be produced with whatever document formatting approach you choose, but what is uploaded must be in PDF. You should upload this to the canvas submission link, before the due date. The report should begin with a cover sheet, which follows the template we provide, and identifies the member who produced this report.

Part A: An extended ER diagram, which was checked by the member for validity. This need NOT be the final diagram submitted in the group report; indeed, it will most likely be a preliminary version which the group considered earlier in the design process. Note that there are NO marks here for the quality of the design here; this is just so the marker can check the statements made in Part B below.
Part B: A description of how the member checked the diagram in Part A. You

should indicate which phrases from the textual description you checked, and what you found in the diagram that captures this (or does not fully capture it). Also indicate aspects of the diagram which you checked by seeing how the text expresses this, and whether or not it matches. Your discussion should include an overview of strengths and limitations of the conceptual model, and any suggestions you made for changes to the diagram, based on what you found in checking.

Part C: A description of the process the group followed to produce its conceptual model and report; this should state which member did what tasks and when, how the work was checked, how disagreements were resolved, etc. It ought to indicate what was effective and what was ineffective in the process and suggest ways the process could be improved.

How to submit your work

There is a submission link in Canvas for the **group's PDF report**. The file can be uploaded by any member of the group, and all members should then see it as a submission they made. You can resubmit at any time up to the due date; we will mark the last submission before the due date. Once the work is marked, the mark will be shown.

There is a submission link in Canvas for you to upload the **individual PDF report**. You can resubmit at any time up to the due date; we will mark the last submission before the due date. Once the work is marked, the mark will be shown.

Marking

Marks will be posted in Canvas once marking is finished for the class.

Group report (worth 5 points):

Part A is worth 3 points. This reflects how well the conceptual data model shown in the EER diagram, captures the described information and uses the notation correctly.

Part B is worth 2 points. The mark here is for how well the discussion identified aspects where ambiguity needed to be resolved, and of strengths and limitations of the conceptual model in relation to the text (e.g. cases where constraints could not be shown directly)

Individual report (worth 7 points):

Part A is not marked

Part B is worth 4 points. This reflects how thoroughly and sensibly the report describes checks on the conceptual model shown in Part A; especially, whether the checks cover a wide variety of aspects of the domain and of the conceptual model, and whether the description reaches appropriate conclusions about the strengths or limitations of the conceptual model.

Part C is worth 3 points. The mark is for how well the report conveys the process that was followed by the group (including when things happened and who did them), and how insightful are the reflections on strengths and limitations of the process, along with suggestions for improving it.